

FUNDAMENTALS OF C

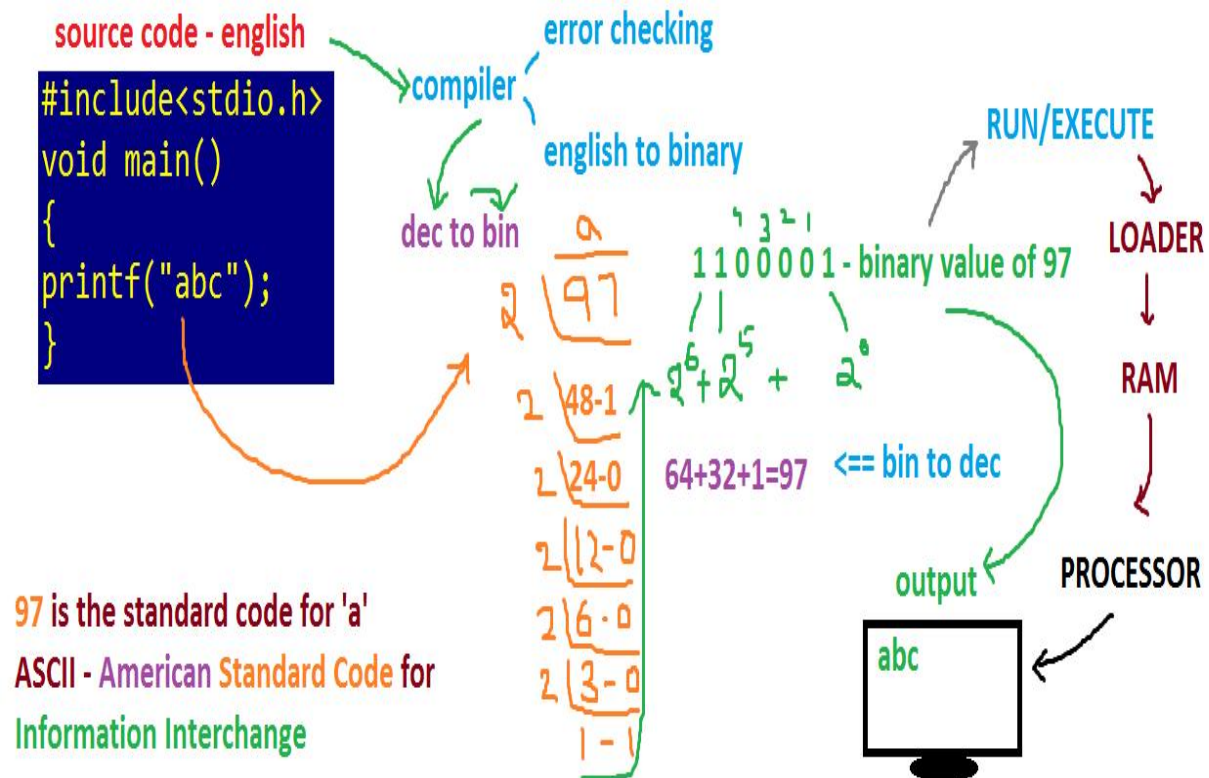
C CHARACTER SET: every programming language comes with a particular character set and by using this characters set only we can write the programs [software]. C uses ASCII character set which comes with 256 characters. In this **52 alphabets** [a-z, A-Z], **10 digits** [0-9], **44 operators** [+,-, *,...], **14 separators** [, . “ “ ‘ ‘ ; :] and **remain all are special characters.**

English language	C language
26 alphabets	256 ASCII values
English words	Keywords - 32
English sentences	Instructions
English paragraphs	Programs
English documents	software

ASCII – American standard code for information interchange – 256 – IBM [International business machines]

Java / .net / python – UNICODE [Universal code] – 65536 characters

Compilation and execution phases:



CHARACTERS	ASCII VALUES
a-z	97-122
A-Z	65-90
0-9	48-57
Space	32
*	42
+	43
Back space	8

Tab key	9
Enter	10
Esc	27
\0	0

C TOKENS

The smallest individual words we are using in developing a c program are called tokens. They are of different types.

1. **Keywords:** The system predefined / reserved words are called keywords. Each keyword is having certain meaning and as a user we can't change this meaning. C comes with 32 keywords.

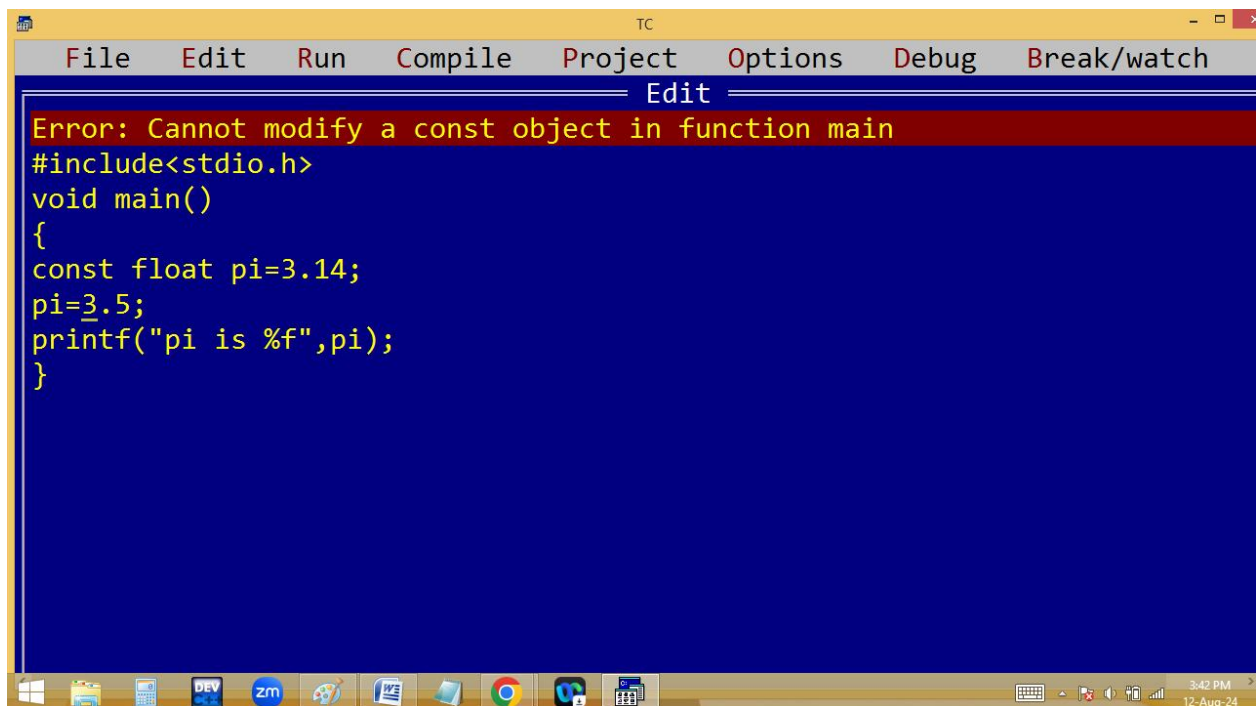
Eg: auto, break, continue, char, const, case, do, default, double, else, enum, extern, for, float, long, int, while, switch, register, structure, union, typedef, signed, unsigned, short,

2. **Constants:** Fixed values. We can't change the constant value during program execution.

Constant value should be provided at the time of declaration only. i.e. further initializations not allowed.

Eg:

const float pi = 3.14;



The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The 'Edit' menu is currently open. A red error message banner at the top of the code editor reads: "Error: Cannot modify a const object in function main". Below the banner, the following C code is visible:

```
#include<stdio.h>
void main()
{
const float pi=3.14;
pi=3.5;
printf("pi is %f",pi);
}
```

 The code attempts to declare a constant float variable 'pi' and then modify its value, which causes the error. The Windows taskbar at the bottom shows various application icons and the system clock indicating 3:42 PM on 12-Aug-24.

